

PAPER_377 — compute_a_wormhole() Implementation & MUGE Safety Infrastructure

Author: Daniel T. Murphy **Date:** 2025

Source: grok_share_11254865.txt, lines 8600-10322 (C++ v8 and v9 final programs)

Session: 102 (final unread block, confirmed from complete file read) **CP4 Class:** Worm-holeMUGETermImplSafetyCalculator (CP4 #26)

Abstract

$$F_{U,Bi} = \kappa \cdot \frac{\rho_{SCm}}{\rho_{UA}} \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_m + U_{bi})$$

This paper presents a UQFF analysis of compute_a_wormhole() Implementation & MUGE Safety Infrastructure, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

This paper captures the formal C++ implementation of the wormhole coupling term in the Resonance MUGE framework, along with division-by-zero error safety in the Compressed MUGE functions, a complete 24-test assertion suite, and full CSV I/O for MUGESystem data. These form the production-ready computational layer for the UQFF framework.

2. compute_a_wormhole() — Implemented Function

Canonical Form

```
double compute_a_wormhole(double r, double b = 1.0,
                           double f_worm = 1.0,
                           double Evac_neb = 7.09e-36) {
    return f_worm * Evac_neb * (1.0 / (b * b + r * r));
}
```

Physical Meaning

$$a_{\text{worm}} = f_{\text{worm}} \cdot \text{Evac_neb} / (b^2 + r^2)$$

- $b = 1.0 \text{ m}$ — Morris-Thorne throat radius (PAPER_373 baseline)
- $f_{\text{worm}} = 1.0$ — wormhole coupling constant (dimensionless, unit default)
- $\text{Evac_neb} = 7.09 \times 10^{-36} \text{ J/m}^3$ — nebular vacuum energy density
- Denominator $(b^2 + r^2)$ — wormhole geometry: $r \rightarrow 0$ gives throat maximum, $r \rightarrow \infty \rightarrow 0$

Numerical values:

At $r = 1e4 \text{ m}$, $b = 1.0$: $a_{\text{worm}} = 7.09e-36 / (1 + 1e8) \approx 7.09e-44 \text{ m/s}^2$

At $r = 1e12 \text{ m}$, $b = 1.0$: $a_{\text{worm}} = 7.09e-36 / 1e24 = 7.09e-60 \text{ m/s}^2$

At $r = 0.0 \text{ m}$, $b = 1.0$: $a_{\text{worm}} = 7.09e-36 / 1.0 = 7.09e-36 \text{ m/s}^2$ (throat max)

Integration into compute_resonance_MUGE()

```
double compute_resonance_MUGE(const MUGESystem& sys, const ResonanceParams& res) {  
    double aDPM          = compute_aDPM(sys, res);  
    double aTHz           = compute_aTHz(aDPM, sys, res);  
    double avac_diff      = compute_avac_diff(aDPM, sys, res);  
    double asuper_freq    = compute_asuper_freq(aDPM, res);  
    double aaether_res    = compute_aaether_res(aDPM, res);  
    double Ug4i           = compute_Ug4i(aDPM, sys, res);  
    double aquantum_freq  = compute_aquantum_freq(aDPM, res);  
    double aAether_freq   = compute_aAether_freq(aDPM, res);  
    double afluid_freq    = compute_afluid_freq(sys, res);  
    double Osc_term       = compute_Osc_term();  
    double aexp_freq      = compute_aexp_freq(aDPM, sys, res);  
    double fTRZ           = compute_fTRZ(res);  
    double a_worm         = compute_a_wormhole(sys.r);    // ← NEW final term  
  
    return aDPM + aTHz + avac_diff + asuper_freq + aaether_res  
        + Ug4i + aquantum_freq + aAether_freq + afluid_freq  
        + Osc_term + aexp_freq + fTRZ + a_worm;    // 13 total terms  
}
```

3. Error-Safe Compressed MUGE Functions

Division-by-zero protection added to 4 compressed MUGE functions:

```
double compute_compressed_base(const MUGESystem& sys) {  
    if (sys.r == 0.0)  
        throw std::runtime_error("Division by zero in r");  
    return G * sys.M / (sys.r * sys.r);  
}  
  
double compute_compressed_super_adj(const MUGESystem& sys) {  
    if (sys.Bcrit == 0.0)  
        throw std::runtime_error("Division by zero in Bcrit");  
    return 1 - sys.B / sys.Bcrit;  
}  
  
double compute_compressed_quantum(double hbar, double Delta_x_p, ...) {  
    if (Delta_x_p == 0.0)  
        throw std::runtime_error("Division by zero in Delta_x_p");  
    return (hbar / Delta_x_p) * integral_psi * (2 * PI / tHubble);  
}  
  
double compute_compressed_perturbation(const MUGESystem& sys) {  
    if (sys.r == 0.0)  
        throw std::runtime_error("Division by zero in r^3");  
    return (sys.M + sys.M_DM) * (sys.delta_rho_rho + 3*G*sys.M / (sys.r*sys.r*sys.r));  
}
```

4. Complete 24-Test Assertion Suite

The final test suite (run_unit_tests()) contains 24 tests:

Test Function	Expected Value	Source
test_compute_compressed_base	$6 \times M_{\text{sun}} / (1\text{AU})^2 \approx 0.0059$	Newtonian validation
test_compute_compressed_expansion	1.0 (at t=0)	Zero-time boundary
test_compute_compressed_superficial	$0.9 \text{ (at } B_i = 1\text{e}10, B_{\text{crit}} = 1\text{e}11)$	$B/B_{\text{crit}} = 0.1$
test_compute_compressed_fluid	$4.1189\text{e-}2$	$\rho \times V \times g$ product
test_compute_compressed_env	1.0	Identity
test_compute_compressed_Ug	0.50m	Simplified
test_compute_compressed_cosm	$1\text{e-}52 \times c^2/3$	Λ constant
test_compute_compressed_quantum	$(h/1.68) \times 2.176\text{e-}18 \times (2\pi/4.35\text{e}17)$	Hubble time
test_compute_compressed_perimeter	$M_{\text{vir}} \times (1.5 + 3GM/r^3)$	SGR1745 params
test_compute_compressed_MUG	$7.82\text{e}39 \text{ m/s}^2$	SGR1745 vs document
test_compute_aDPM	$3.545\text{e-}42 \text{ m/s}^2$	SGR1745
test_compute_aTHz	$1.182\text{e-}33 \text{ m/s}^2$	aDPM= $3.545\text{e-}42$, vexp= $1\text{e}3$
test_compute_avac_diff	$3.545\text{e-}53 \text{ m/s}^2$	aDPM= $3.545\text{e-}42$, vexp= $1\text{e}3$
test_compute_asuper_freq	$1.048\text{e-}21 \text{ m/s}^2$	aDPM= $3.545\text{e-}42$
test_compute_aaether_res	$3.900\text{e-}38 \text{ m/s}^2$	aDPM= $3.545\text{e-}42$
test_compute_Ug4i	0.0 (Ereact $\approx$ 0 at t= $3.799\text{e}10$)	Decay asymptote
test_compute_aquantum_freq	$1.708\text{e-}66 \text{ m/s}^2$	Hubble resonance scale
test_compute_aAether_freq	$1.863\text{e-}84 \text{ m/s}^2$	Aether freq
test_compute_afluid_freq	$1.773\text{e-}9 \text{ m/s}^2$	SGR1745 afluid confirmation
test_compute_Osc_term	0.0	Identity
test_compute_aexp_freq	$1.623\text{e-}57 \text{ m/s}^2$	SGR1745 at t= $3.799\text{e}10$
test_compute_fTRZ	0.1	Default value
test_compute_resonance_MUG	$7.73\text{e-}9 \text{ m/s}^2$	SGR1745 total resonance
test_compute_a_wormhole	$1/(1+r^2)$ (at Evac_neb=1, b=1)	NEW in v9

```
void test_compute_a_wormhole() {
    double r = 1e4;
    double b = 1.0;
    double expected = 1.0 / (1.0 + r * r); // f_worm=1, Evac_neb=1 for test
    double result = compute_a_wormhole(r, b, 1.0, 1.0);
    assert(std::abs(result - expected) < 1e-6);
}
```

5. CSV File I/O — load_muge_systems()

Complete 18-field CSV parser for MUGESystem:

```
std::vector<MUGESystem> load_muge_systems(const std::string& filename) {
    std::vector<MUGESystem> systems;
    std::ifstream in(filename);
    if (in.is_open()) {
```

```

std::string line;
while (std::getline(in, line)) {
    std::stringstream ss(line);
    MUGESystem sys;
    std::string token;
    std::getline(ss, sys.name, ',');
    std::getline(ss, token, ','); sys.I = std::stod(token);
    std::getline(ss, token, ','); sys.A = std::stod(token);
    std::getline(ss, token, ','); sys.omega1 = std::stod(token);
    std::getline(ss, token, ','); sys.omega2 = std::stod(token);
    std::getline(ss, token, ','); sys.Vsys = std::stod(token);
    std::getline(ss, token, ','); sys.vexp = std::stod(token);
    std::getline(ss, token, ','); sys.t = std::stod(token);
    std::getline(ss, token, ','); sys.z = std::stod(token);
    std::getline(ss, token, ','); sys.ffluid = std::stod(token);
    std::getline(ss, token, ','); sys.M = std::stod(token);
    std::getline(ss, token, ','); sys.r = std::stod(token);
    std::getline(ss, token, ','); sys.B = std::stod(token);
    std::getline(ss, token, ','); sys.Bcrit = std::stod(token);
    std::getline(ss, token, ','); sys.rho_fluid = std::stod(token);
    std::getline(ss, token, ','); sys.g_local = std::stod(token);
    std::getline(ss, token, ','); sys.M_DM = std::stod(token);
    std::getline(ss, token, ','); sys.delta_rho_rho = std::stod(token);
    systems.push_back(sys);
}
}
return systems;
}

```

CSV Format (18 fields):

name,I,A,omega1,omega2,Vsys,vexp,t,z,ffluid,M,r,B,Bcrit,rho_fluid,g_local,M_DM,delta_rho_rho

6. Command-Line I/O

```

int main(int argc, char** argv) {
    std::string input_file;
    std::string output_file;
    for (int i = 1; i < argc; i += 2) {
        std::string arg = argv[i];
        if (arg == "--input" && i + 1 < argc) input_file = argv[i + 1];
        if (arg == "--output" && i + 1 < argc) output_file = argv[i + 1];
    }
    // If --input given: load_muge_systems(input_file)
    // If --output given: redirect std::cout to file
}

```

7. Multi-File Architecture (Proposed)

Header decomposition for modular build:

main.cpp – command-line entry, orchestration

celestial.h/cpp – CelestialBody struct, Ug1/Ug2/Ug3/Ug4/Um/FU functions
 muge.h/cpp – MUGESystem, ResonanceParams, compressed+resonance MUGE
 fluidsolver.h/cpp – FluidSolver (Navier-Stokes), simulate_quasar_jet

CMakeLists.txt:

```
cmake_minimum_required(VERSION 3.10)
project(StarMagic)
set(CMAKE_CXX_STANDARD 11)
add_executable(star_magic main.cpp celestial.cpp muge.cpp fluidsolver.cpp)
```

8. Key Numerical Reference (Wormhole term across 7 systems)

System	r (m)	a_worm (m/s ²)
Magnetar SGR 1745-2900	1e4	7.09e-36 / 1e8 $\approx$ 7.09e-44
Sagittarius A*	1e12	7.09e-36 / 1e24 $\approx$ 7.09e-60
Tapestry of Blazing Starbirth	3.086e17	7.09e-36 / 9.52e34 $\approx$ 7.44e-71
Westerlund 2	3.086e17	same as Tapestry
Pillars of Creation	9.46e15	7.09e-36 / 8.95e31 $\approx$ 7.92e-68
Rings of Relativity	3.086e17	7.09e-36 / 9.52e34 $\approx$ 7.44e-71
Student's Guide	1e26	7.09e-36 / 1e52 $\approx$ 7.09e-88

The wormhole term is always subdominant ($\ll$ other MUGE terms), confirming it acts as a geometrically-grounded perturbation rather than a dominant contribution.

9. CP4 Class

Class: WormholeMUGETermImplSafetyCalculator **Category:** Physics Implementation / Validation Infrastructure **References:** PAPER_373 (Morris-Thorne), PAPER_375 (a_worm formula suggestion)

Watermark: ©2025 Daniel T. Murphy, daniel.murphy00@gmail.com – All Rights Reserved

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

 §B.
 VDS/DVP/BSH
 Deep
 Syn-
 the-
 sis

 §B.1
 Vac-
 uum
 Den-
 sity
 Se-
 ries
 (VDS)
 The
 canon-
 ical
 VDS
 ratio

$$\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$$

gov-
 erns
 the
 double-
 exponential
 vac-
 uum
 con-
 den-
 sate
 pro-
 file:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\text{VDS}}}\right)\right)$$

##

§A.

Cosmogenesis-

Linked

La-

grangian

(PA-

PER_877

Sym-

bolic

Ex-

port)

For

this

sys-

tem,

the

local

VDS

sub-

ratio

is

0.057

(near-

threshold

regime),

plac-

ing

it in

the

$t \rightarrow$

π

col-

lapse

zone

where

the

double-

exponential

tran-

si-

tions

sharply

from

con-

densed

to di-

lute

vac-

uum.

This

thresh-

old

be-

hav-

io7

con-

nects

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

 §B.2
 Dipole
 Vor-
 tex
 Primes
 (DVP)
 The
 DVP
 en-
 cod-
 ing
 maps
 the
 sys-
 tem's
 char-
 ac-
 teris-
 tic
 pa-
 ram-
 eter
 onto
 the
 prime
 lat-
 tice:
 $p_{\text{DVP}} =$
 $61, \quad n_{\text{channel}} =$
 $14/26$

##

§A.

Cosmogenesis-

Linked

La-

grangian

(PA-

PER_877

Sym-

bolic

Ex-

port)

Since

$p_{\text{DVP}} =$

61 is

res-

o-

nant

(thresh-

old

at

$p >$

26),

the

sys-

tem's

vac-

uum

topol-

ogy

in-

her-

its

reso-

nant

en-

hance-

ment

from

the

DVP

lat-

tice,

am-

plify-

ing

UQFF

cou-

pling

at

spe-

cific

radii

where

com-

pressed

mat-

ter

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

 §B.3
 Buoy-
 ancy
 Sat-
 ura-
 tion
 Har-
 mon-
 ics
 (BSH)
 The
 BSH
 satu-
 ra-
 tion
 timescale
 for
 this
 sec-
 tor
 is
throat
cross-
ing
time
 (geodesic
 sta-
 bi-
 liza-
 tion):
 $\mathcal{F}_{\text{BSH}} =$
 $\sum_{j=1}^{26} \frac{1}{j}.$
 $f_{U_b} \cdot$
 $(1 - e^{-[SSq] \cdot m/M_{\odot}}).$
 $\cos\big(\frac{2\pi j}{26}\big)$

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

The
 tanh
 satu-
 ra-
 tion
 en-
 ve-
 lope
 pre-
 vents
 un-
 phys-
 ical
 di-
 ver-
 gence:
 $\mathcal{F}_{\text{BSH,sat}} =$
 $\mathcal{F}_{\text{BSH}}$
 $\left(1 - \tanh\left(\frac{t-t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

 §A.
 Cosmogenesis-
 Linked
 La-
 grangian
 (PA-
 PER_877
 Sym-
 bolic
 Ex-
 port)

connecting
 to
 the
 PA-
 PER_877
 Stage
 5
 buoy-
 ancy
 seed

$$U_{b,\text{seed}} = 0.1 \cdot (\hbar c / r^2).$$

f_{SCm}
 which
 ini-
 tial-
 izes
 the
 har-
 monic
 se-
 ries
 at
 cos-
 mo-
 gen-
 esis.

 §B.4
 Production-
 Scale
 Con-
 sis-
 tency

§A.
Cosmogenesis-
Linked
La-
grangian
(PA-
PER_877
Sym-
bolic
Ex-
port)

|
Frame-
work
|
Canon-
ical
Value

|
This
Pa-
per |
Sta-
tus |
|—

—
|—
—
—|—
—

—|—
—||

VDS
ratio
|
 $\rho_{\text{SCm}}/\rho_{\text{UA}} =$
1.894

| Lo-
cal
sub-
ratio
=
0.057

| ✓
Threshold-
consistent

||
DVP
prime
|

$p_k \in$
 $\{2,3,...,113\}$
|
 $p_{\text{DVP}} =$
613
✓

Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.